

Fight Log — Append Rows to CSV

Day 04

Lesson 04

Nora XIE

Course 02

Phase 1 — Game Studio

Time: 60–75 min

Mode: Self-study

Today's goals:

- ★ Append one row per fight to fights.csv
- ★ Include round, command, damage, winner
- ★ Keep headers on the first write only



Course 02 quest map: Day 04 of 18

Hook

Studios track what players do. Your first data pipeline starts today — one row per fight.

Learn



Learn

CSV = comma-separated table. Header row first:
round,command,player_hp,dragon_hp,note
Append mode: `open("fights.csv", "a")`.
Use the `csv` module so commas in text do not break the file.
This log becomes your dataset for Phase 2!



AI rules today

Predict/Run: AI off. **Investigate:** Socratic hints only. **Make:** you review every line.

Predict — Run — Investigate — Modify —> Make



Demo code

Folder: day04/

```
1 import csv
2 from pathlib import Path
3
4 LOG = Path("fights.csv")
5
6 def log_fight(round_num, command, player_hp, dragon_hp, note=""):
7     new_file = not LOG.exists()
8     with open(LOG, "a", newline="") as f:
9         w = csv.writer(f)
10        if new_file:
11            w.writerow(["round", "command", "player_hp", "dragon_hp", "note"])
12        w.writerow([round_num, command, player_hp, dragon_hp, note])
```

Full file: demos/demo_csv_log.py

 Try it

In terminal, `cd course02/day04`, then predict output, then run: `python demos/demo_csv_log.py`

 Assignment

Work in: `day04/assignment/`

A — Predict Does mode "a" erase old rows or add new ones?

B — Debug `assignment/buggy_csv.py` — writes header every time.

C — Level up Log every fight and poison in your game to `assignment/fights.csv`.

D — Explain Name two columns you added and why they help analysis later.

 Checklist

- ☐ Re-read **Learn**; predict outputs (10 min)
- ☐ Assignments A–D (35–45 min)
- ☐ Run your program 3 times with different inputs
- ☐ Stuck 15 min → C-R-A-V (`../shared/prompts.md`)

Reflection

1. Easiest part today?
2. One line you can teach a friend.

Tomorrow: Tiny tests so poison never goes negative.